classmate
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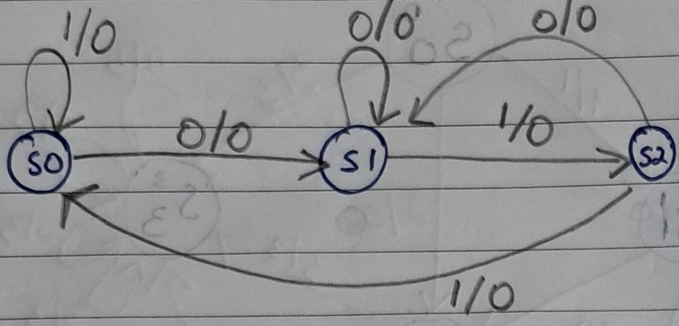
FSMs - Assignment

1.) Draw the state diagram for a FSM, the output toggles whenever it detects '110' LSB first (overlapping). Consider initial value of output as "0".

= State diagram



Previous state		input	next state		output (y)
q_1	q_0	I	D_1	D_0	y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	1
x	x	x	x	x	0
x	x	x	x	x	0

S0 = rows 1–2, S1 = rows 3–4, S2 = rows 5–6.

D_1 =

q_1 \ $q_0 I$	00	01	11	10
0	0	0	1	0
1	0	0	X	X

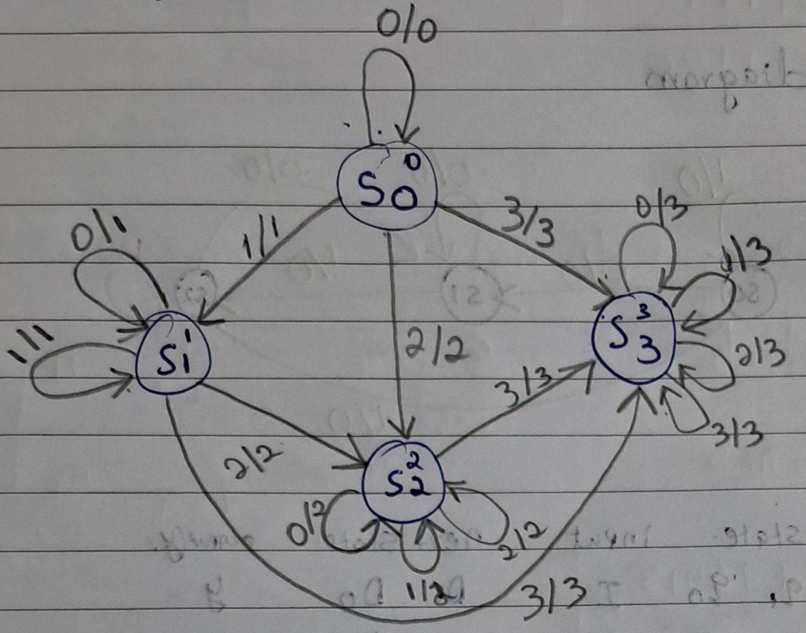
D_0 =

q_1 \ $q_0 I$	00	01	11	10
0	1	0	0	1
1	1	0	X	X

$\therefore D_1 = q_0 I$ $\qquad D_0 = \bar{I}$ $\qquad y = q_1 \bar{q_0} I$

2. Design a FSM over {0,1,2,3} that will output the largest input digit read so for. eg. input : 001032 will produce output : 001133

= State diagram.



Previous State	Input	nextstate
S0	0	S0
S0	1	S1
S0	2	S2
S0	3	S3
S_1	0	S1
S_1	1	S_1
S_1	2	S2
S_1	3	S3
S_2	0	S2
S_2	1	S2
S_2	2	S2
S_2	3	S3
S_3	0	S3
S_3	1	S3
S_3	2	S3
S_3	3	S3